

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2024

Course: B. Tech. Branch: Information Technology Semester: III

Subject Code & Name: BTITC305 Data Structures and Applications

Max Marks: 60

Date: 09/07/2024

Duration: 3 Hrs.

Instructions to the Students:

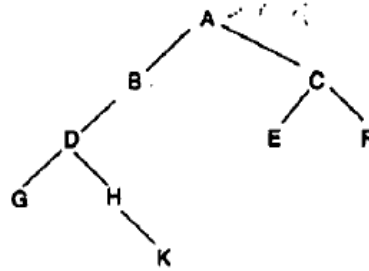
- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What is the data structure? Define an array and a linked list.	Understand Knowledge CO3	6
B) Describe the most frequently used operations on the data structure. Calculate time complexity of the linear search algorithm.	Knowledge Apply CO3	6
C) Write an algorithm: i)To find the solutions of the quadratic equation ii)To find the smallest element in the list	Apply CO1	6
Q.2 Solve Any Two of the following.		12
A) Evaluate the following postfix expression using stack. 12, 7, 3, -, /, 2, 1, 5, +, *, +	Evaluate CO2	6
B) What is a priority queue? Describe a one-way list representation of a priority queue.	Knowledge Understand CO1	6
C) Design a procedure which traverses a linked list, applying an operation PROCESS to each element of a list.	Create CO1	6
Q. 3 Solve Any Two of the following.		12
A) State the principle of inorder traversal of binary tree. Illustrate with an example.	Knowledge Apply CO3	6

- B) Write a stack based algorithm for preorder traversal of a binary tree.
Apply it on the following binary tree, show the steps.

Create
Apply
CO3

6



- C) Define binary search tree.
Design an algorithm which finds location of the given item in a binary search tree or inserts it as a new node at its appropriate place.

Knowledge
Create
CO1

6

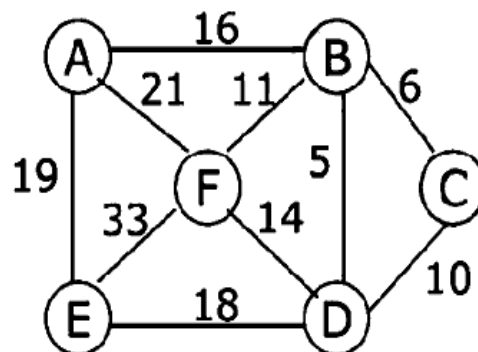
Q.4 Solve Any Two of the following.

12

- A) Describe the following terms with reference to Graph.
i) Spanning Tree
ii) Adjacency List
iii) Path Matrix
- B) Find the minimum spanning tree using kruskal's algorithm for the following graph.

Understand
CO1

6



- C) Explain sequential representation of a graph with a suitable example.

Understand
CO3

6

Q. 5 Solve Any Two of the following.

12

- A) Write a C program to implement a Linear Search algorithm for searching a given item in the list.
- B) Sort the following list using Insertion Sort, show the passes.
77, 33, 44, 11, 88, 22, 66, 55
- C) Write a C program to implement a Selection Sort algorithm for sorting a given list.

Create
CO4
Apply
CO3
Create
CO5

6

6

6

*** End ***